

55th CIRP Conference on Manufacturing Systems Guideline for Deployment of Machine Learning Models for Predictive Quality in Production

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Abstract

Predicting product quality represents a common area of application of machine learning (ML) in manufacturing. However, manifold challenges occur during the integration of ML models into production processes. Therefore, this paper aims to provide a guideline for the deployment of ML models in production environments. Relevant decisions and steps for deploying models in predictive quality use cases are demonstrated. The results for each component of the proposed guideline – deployment design, productionizing & testing, monitoring, and retraining – have been validated with industry experts including exemplary implementations.

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1. Introduction

In a globalized world, companies from industrialized nations must be able to manufacture products of superior quality at competitive costs [1]. One way of achieving the highest possible level of quality is using ML to analyze past defects in production and predict future events to derive remedial measures [2]. ML as a subdomain of artificial intelligence (AI) is an emerging subject in investigation with increasing interest from the industry [3].

Deployment is necessary for making use of performant ML models in production. Studies and surveys from practice show that a high percentage of ML projects fail without a successful deployment [4]. As a consequence of unsuccessful deployments, companies waste a substantial amount of resources [5]. Due to numerous technological but also organizational challenges, deploying ML models into the running production

process proves to be enormously difficult [6]. From a technological point of view, a main challenge consists in the constantly changing landscape of software tools for data and AI applications [7]. At the same time, the support and understanding by decision owners as well as the alignment of ML with the business problem are crucial aspects to overcome the organizational barriers for deployment [8].

This paper aims to provide a framework for the ML model deployment applied to the context of predictive quality in production. A comprehensive guideline is developed to facilitate the successful integration of ML models for predictive quality. On the one hand, it can be applied to a wide range of scenarios due to its generic approach. On the other hand, the concept considers the individual requirements of predictive quality use cases by providing relevant decisions and steps. The deployment is broken down in a structured and target-oriented manner to provide the best value for decision owners with

background in business or production with no expert knowledge in software or ML.

The remainder of this paper is structured as follows. Section 2 describes the procedure of the literature review and presents the key findings. In section 3, the methodology for the guideline development is presented followed by the results in section 4. How the guideline can be applied to a predictive quality use case in the production context is described in section 5. Section 6 provides concluding remarks.

2. Literature Review

Within the Cross Industry Standard Process for Data Mining (CRISP-DM), deployment is understood as the integration of the ML model into an organization's decision-making processes [9]. In this paper, the focus is set on deploying ML models for predictive quality. Predictive quality is defined as “the empowerment of the user to optimize product and process-related quality by using data-driven predictions as a basis for decision-making and action” [10]. For the literature review, publications are accumulated, out of which the most relevant ones are then selected. Subsequently, the selected publications are categorized and evaluated.

In accordance with the fact that the topic of deployment of ML models for predictive quality in production is located on the verge of multiple fields of investigation, search strings from ML, production and software engineering are used in combination. Gray literature is also considered for the search as academic literature only gives an incomplete view on dynamic fields of investigation [11]. Publications by practitioners represent valuable sources of information and may have high relevance for the topic. Nonetheless, they require a profound analysis of worthiness before being selected.

A categorization allows classifying the selected items based on their origin either in production engineering or software engineering. Subsequently, selected approaches are evaluated by means of four criteria illustrated in Fig. 1.

ML for predictive quality	1	3	Guideline structure
Deployment of ML	2	4	Transferability

Fig. 1. Evaluation criteria.

First, it is analyzed how detailed the use of *ML for predictive quality* (criterion 1) is treated. Similarly, criterion 2 evaluates the covered depth of the topic *deployment of ML*. Furthermore, it is assessed to which degree a *guideline structure* is followed (criterion 3). The fourth and last criterion evaluates the *transferability* of presented concepts to other use cases (criterion 4). Table 1 gives an overview of the categorization and evaluation of existing approaches by means of the introduced set of criteria.

Table 1. Evaluation results.

Source	ML for predictive quality	Deployment of ML	Guideline structure	Transferability
<i>Production focus</i>				
[12]	●	○	○	●
[13]	●	●	○	●
[14]	●	●	○	●
[15]	●	●	○	○
<i>Software focus</i>				
[16]	○	●	●	●
[17]	○	●	●	●
[18]	○	●	●	●
[19]	○	●	●	●
[20]	○	●	●	●
[21]	○	●	○	●
[22]	○	●	○	●

● completely fulfilled ● partly fulfilled ○ not fulfilled

As an excerpt of the overall evaluation results, three relevant publications are highlighted and presented in more detail at this point. Schmitt J et al. [14] provide a new integrated solution of predictive model-based quality inspection in industrial manufacturing using ML techniques and Edge Cloud Computing technology. In the provided framework, deployment and the technical implementation into an existing IT plant infrastructure are covered from a high-level perspective. Even though transferability is given, the approach does not point out different deployment options.

Sato et al. [18] introduce the concept of Continuous Delivery for Machine Learning, which brings continuous delivery principles and practices to ML applications. In the form of a guideline, an end-to-end lifecycle from model building to deployment and monitoring is presented. Due to the high level of detail provided by the well-respected practitioners, the approach is very frequently cited within the community. However, no connection to industrial production or quality-related applications in manufacturing is established.

John et al. [22] conduct a systematic literature review of the state of the art and practice of architecting AI deployment. A framework about how to integrate, deploy, operationalize, and evolve AI models is derived for the end-to-end deployment process. Relevant practices and challenges are extracted from literature and compared creating a comprehensive overview of necessary, well-transferable tasks. Given its origin in software, the approach does not address characteristics of predictive quality use cases in production environments.

Table 1 and the preceding exposition show that no existing approach fulfills all criteria sufficiently and the overall goal of the paper is not met. Therefore, further research is necessary.

3. Methodology

Based on the identified research gap, a guideline for the deployment of ML models for predictive quality is developed. In prior research regarding ML pipelines in production, the blocks data integration, data preparation, modeling and

deployment are defined [12, 23]. In the proposed guideline, the deployment is divided into deployment design, productionizing & testing, monitoring, and retraining. There are dependencies between the components in form of sequential execution and feedback loops. Since the deployment design defines the system's design requirements, it serves as an input for the subsequent steps and remains unchanged during the process (Fig. 2).



Fig. 2. Deployment guideline components.

For each component, necessary decisions and steps for a successful deployment are identified based on existing approaches from theory paired with insights from the practical validation. Methods used for validation are described in a subsequent section. To develop the guideline, concepts from different sources are aligned and brought together in a clear structure. While doing so, a strong focus is set on predictive quality applications.

At the end of the development process, there is a comprehensive guideline for deploying ML models in predictive quality use cases. Nonetheless, the underlying structure of the developed procedure is sufficiently generic so that it is extendable to further use cases such as predictive maintenance or anomaly detection. In alignment with the overall objective of the paper, the detailed description of decisions and steps ensures a systematic approach during deployment.

4. Results

In the following, each guideline component is explained concerning the application in predictive quality. Furthermore, collaboration and tool selection are covered as overarching aspects relevant for fruitful deployment efforts.

4.1. Deployment design

As the goal of the deployment design, requirements towards the ML deployment system architecture are determined. For this purpose, five design parameters in conjunction with the respective options are identified (Fig. 3).

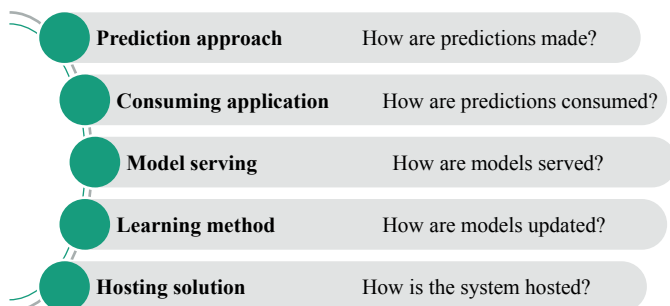


Fig. 3. Deployment design parameters.

By choosing one value for each parameter in form of a morphological box, existing architectures can be replicated, and new, unseen ones can be created based on the use case characteristics. Applying the concept of batch and streaming processing to predictive quality, predictions about the product quality can be calculated on batches of input data or in real time [13]. Time critical applications such as real-time parameter adjustments are treated differently to scenarios where there is a time buffer between the generation of input data and the use of the model's outputs. Predictions can be consumed in a web app, which can be accessed in the browser or by other services to retrieve the quality predictions for purposes of further analysis such as dashboards. Alternatively, a native app is installed on a smart device or integrated into the machine control software to process the predictions. The model can be embedded into the consuming application in a monolith architecture or served as a separate service to achieve a microservice architecture. In the case of a separate model serving, the prediction results delivery can be realized via database, REST API or streaming [18, 22]. In accordance with the modeling phase before the deployment, the model can learn offline on batches of training data or online with every new data point [24]. Options for hosting the overall system include on-premises such as in-device or on-site as well as in a public or private cloud [21]. Which option to select for each parameter depends on the circumstances of every individual case. Existing boundary conditions heavily influence the decision. Furthermore, effects on the complexity and scalability of the target system must be considered.

4.2. Productionizing & testing

Productionizing is understood as a series of implementation tasks. Testing ensures reproducibility during an ML model's transfer from the research environment to a production environment. In this context, the term production is used ambiguously for the final stage of software development as well as for the industrial production as in manufacturing and assembly. This component aims to develop a deployable application around the ML model. As a result, this leads to a process that is similar to regular software development (Fig. 4).

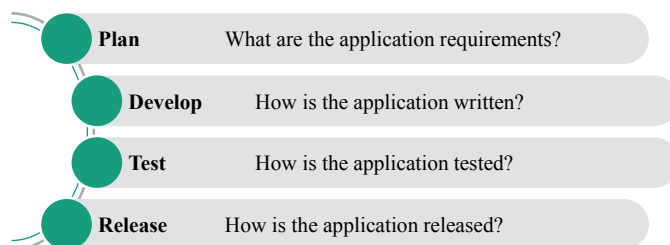


Fig. 4. Productionizing & testing process.

Nonetheless, in the case of ML the process is more complex than for traditional software. In addition to the code, there are two more ML-specific dimensions in form of data and model [18]. This operational process is generally valid and not limited to predictive quality use cases but represents one of the most time-consuming activities in deployment.

First, the required ML functionality is to be planned, that the application needs to fulfill in an isolated manner or in interaction with other systems. On this basis, the code for the application is written including the transfer of modeling code, e.g., provided by a data scientist in Jupyter Notebook, into the target programming language and creating a serialized object of the ML model respectively multiple models for further use. The deployment of ML models does not only refer to bringing an ML algorithm into production but includes the whole pipeline containing data preparation steps plus the algorithm [25]. In addition to the code itself, training data and resulting models need to be versioned and tracked. Operational and ML-related tests are essential to ensure that only well-performing ML model applications are rolled out to the users. Different release strategies exist to safely replace a running application with a newer version. To increase speed and reliability of the productionizing & testing process, the phases can be automated in the sense of Continuous Delivery [18].

4.3. Monitoring

Once in production, the ML model application is monitored on different levels. In this guideline, the monitoring is analyzed from an ML perspective rather than from an operational point of view (Fig. 5).

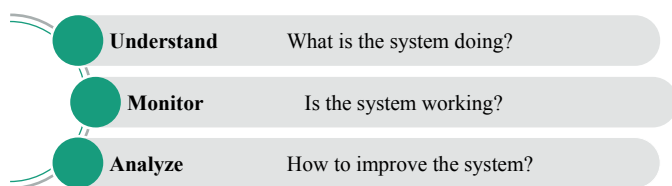


Fig. 5. Monitoring levels.

Metrics, logs and traces are called the three pillars of observability and enable to understand the behavior of a system for predictive quality [26]. The same metrics as in the modeling phase are applied for the monitoring. In the case of binary classification tasks, e.g., classifying a product's quality as ok or not ok, four metrics are relevant: accuracy, precision, recall and F1 score [2]. A model's predictions can be analyzed later by logging input data from production and the predicted quality. If multiple services are involved with handling a prediction request, traces represent a method to follow the request through the system. In the case of microservice architectures, tracing allows to determine exactly where a delay or an error occurred, e.g., in data fetching or data pre-processing.

All gathered monitoring information can be displayed in dashboards as a high-level view for the management. In predictive quality, such dashboards combine two kinds of information. On the one hand, they allow process monitoring to track the product quality predicted by the ML model. On the other hand, they are used for monitoring the performance of the ML model application itself. Alerts are directly addressed notifications for the responsible entity based on predefined criteria such as intervention and warning limits for metrics. On top of that, a root cause analysis is executed to derive adequate measures. Possible root causes include hardware errors such as

broken sensors as well as ML-independent software errors such as code bugs or unavailability of services. In the case of ML model performance issues, a separate analysis aims to detect drifts that lead to a deteriorating model performance [27]. A drift may cause the same sensor values to not lead to the same product quality in predictive quality applications. This can be caused by effects such as tool wear, that are not considered by the ML model.

4.4. Retraining

Retraining of ML models becomes inevitable in order to maintain a high performance [28]. Three interconnected decisions are identified as relevant for the retraining (Fig. 6).

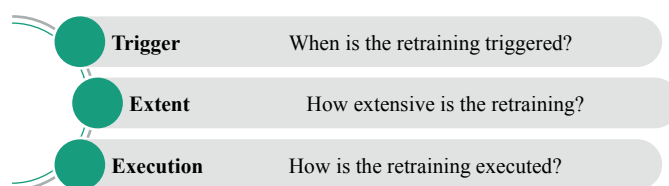


Fig. 6. Retraining decisions.

Periodic triggering of the retraining based on a fixed schedule has the disadvantage of inefficient use of resources [29]. Alternatively, a model can be retrained on demand based on defined criteria. In predictive quality use cases, notifications from monitoring about performance measures falling below a warning or intervention limit facilitate targeted actions. Based on the monitoring data, the extent of the retraining is determined [30]. Retraining in the narrower sense consists of training a model on a new data set. If this does not achieve the desired performance, a redesign of the model is necessary. In that case, a new model pipeline with different data preprocessing steps is set up and trained. Whereas in the retraining only an adjustment of a model's weights is realized e.g., by increasing the feature importance of a particular sensor value, redesign involves adding or removing data sources or sensors, selecting ML algorithms and tuning hyperparameters. Retraining is closely connected to the initial model building and, thus, all decisions must be aligned with the modeling execution. Approaches to reduce the human involvement in the training and retraining through automated ML (AutoML) must be analyzed for every specific given case [28].

Retraining is the last component of the guideline but does not mark the end of the deployment. Rather, productionizing & testing, monitoring, and retraining are understood as a cycle of continuously performed tasks.

4.5. Collaboration and tool selection

Overarching aspects for deploying ML models are the assurance of collaboration and selection of tools. From an organizational point of view, the success of deployment depends on the alignment of three types of expertise in the project: process expertise, data science expertise and DevOps expertise [22]. The inclusion and support of stakeholders from the business perspective of ML is crucial and can be enhanced

through explainability techniques [19]. Comprehending why an ML model assigned a quality label to a product increases the user's trust in the prediction and even allows a better understanding of the process by experts.

The selection of adequate tools poses a critical technological challenge. First, a decision between open source tools and proprietary solutions is required and is followed by the selection of a particular product. Relevant evaluation criteria consist in functional, technical, organizational, legal, economical, and political requirements [31]. The weights of the criteria are adjusted individually in alignment with the use case and the company's characteristics. Regarding companies in the manufacturing industry planning to implement ML for predictive quality, the final selection can vary strongly between small or medium sized enterprises in comparison to large corporations.

5. Validation

Validating conceptual models consists in asking if the developed model contains all necessary details to achieve the study's goals [32]. To ensure that the proposed guideline satisfies its intended use and user needs, it was validated by means of a consortium study with industry partners. Within the International Center for Networked and Adaptive Production (ICNAP) three Aachen-based Fraunhofer Institutes and renowned industrial partners are developing solutions for production systems and value chains aimed at Industry 4.0. Validation was achieved through interviews with experts and an exemplary implementation of a predictive quality use case.

5.1. Expert interviews

Input and feedback of practitioners are valuable sources for the refinement of the guideline. On the one hand, interviews with deployment experts were conducted. All three interviewed experts have a background in research and had deployed ML models in past research and industry projects. As a result, the interviews show that previously realized deployments provided by the experts are covered by the guideline. Additionally, the guideline enables the realization of more efficient deployments in the future by introducing new approaches from the literature review, which have not been fully adapted in the practical deployment yet.

On the other hand, all components were validated in workshops within the ICNAP study through discussions with experts from a corporate background. In each one of the multiple workshops, the guideline was discussed with up to six experts. Whereas all the involved companies operate in the production industry, their sizes range from medium to large enterprises. Likewise, the participants' areas of work vary including the fields of data science, software, and process monitoring. Due to the wide ranges of competences the focus was set on validating the completeness and understandability of the guideline. Based on the experts' feedback, the orientation and elaboration of the guideline were adjusted to the expressed needs.

5.2. Hands-on implementation

Based on the realistic requirements provided by the ICNAP community members, an existing trained ML model was deployed following the guideline from deployment design, productionizing & testing, monitoring, and retraining. A publicly available real-life data set from semiconductor manufacturing provided by the University of California Irvine [33] was used for the initial training. In the use case, instances are classified as pass or fail based on sensor data collected during the process. The deployed ML model application includes prediction, monitoring, and retraining functionality. Wrapped into a Flask backend, the model returns predictions based on requests of data in JSON format. Furthermore, available model versions can be retrieved, monitoring data in form performance metrics or a confusion matrix can be calculated, and new model versions can be trained on request. Using the software Postman, it is possible to make requests to the application's endpoints and visualize responses. Especially the practical implementation could ensure the guideline's applicability for the user.

6. Conclusion

In this paper, a guideline for deploying ML models in production environments was developed. It supports companies to fully reap the benefits of modeling efforts by deploying built models into production and using their results, where they create added value. Given the importance of ensuring successful deployments, a clearly structured methodology was lacking. Therefore, the deployment was broken down into separate guideline components. For each component, relevant decisions and steps were provided in a high-level, generally valid form. Based on this, it was described in detail how introduced decisions and steps can be applied to predictive quality under consideration of individual needs and restrictions of any specific use case.

Limitations exist regarding the area of application. The guideline was conceptualized for predictive quality applications in production. In its current form, it cannot address all deviating scenarios with different requirements towards the deployment. In this context, the validation was also conducted with a limited scope.

Due to the relevance of deployment across industries, future lines of investigation include an analysis of how to adjust the guideline to further use cases within or beyond the manufacturing industry. Furthermore, the integration of DevOps techniques into the ML life cycle with the ultimate goal of automating the whole end-to-end ML pipeline from data integration to deployment is part of future research.

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